

BAALVION INDUSTRIES PRIVATE LIMITED

Global Trade Super-Platform — The World's First End-to-End Trade Operating System

MASTER PROJECT PLAN

Complete Vision | Architecture | All 14 Modules | Step-by-Step Workflow | 350-Prompt Build Plan

\$33T	198	14	9	350	6
Global Market	Countries	Modules	User Portals	Build Prompts	Build Phases

Consolidated from: Baalvion Master Document v4.0 | CTO Master Plan v2.0 | Final 360 Audit v3.0 | Gap Analysis v1.0 | TradeOS CTO Plan v1.0

Version 1.0 | Confidential | 2025 | Baalvion Industries Private Limited

SECTION 1: WHAT EXACTLY WE ARE BUILDING

Baalvion Industries Private Limited is building the world's first end-to-end Global Trade Operating System. This is not a marketplace, not a fintech app, and not a logistics platform. It is all of them — unified into one platform that owns the complete trade journey from the first conversation to the final payment settlement.

The Problem We Solve

HOW GLOBAL TRADE WORKS TODAY	HOW BAALVION WORKS
Find suppliers on Alibaba — unverified, no trust data	Verified sellers with trust scores 0-1000 inside Baalvion Marketplace
Negotiate via email and WhatsApp — no record keeping	In-platform Deal Room — every offer, counter-offer, document locked inside Baalvion
Finance via a separate bank — takes 5-10 days	Letters of Credit, Bank Guarantees, Invoice Finance — same day, inside platform
Book shipping via Flexport separately	Sea, air, road, rail freight comparison and one-click booking inside Baalvion
File customs manually with paper forms	AI-classified HS codes, auto-filed customs declarations for 198 countries
Buy insurance from a separate broker	Cargo insurance quoted and purchased inside Baalvion in minutes
Receive payment via SWIFT — takes 2-5 days	Multi-currency settlement engine — 24 hours, 150+ currencies supported
Check compliance with separate tools	AI KYC, AML monitoring, and Sanctions screening on every entity, every trade

"Every verified buyer and seller who visits Baalvion will never need to go anywhere else. From finding a supplier, to negotiating a deal, to financing it, shipping it, clearing customs, insuring the cargo, and receiving payment — everything happens here. One platform. 198 countries. Zero need for any other tool."

How Baalvion Beats Every Competitor

Platform	Marketplace	Finance	Logistics	Customs	Insurance	198 Countries
Alibaba / Alibaba.com	YES	NO	NO	NO	NO	NO
Flexport	NO	NO	YES	NO	YES	NO
TradeShift / HSBC Contour	NO	YES	NO	NO	NO	NO
Maersk GTD	NO	NO	YES	YES	NO	NO
BAALVION	YES	YES	YES	YES	YES	YES

SECTION 2: PLATFORM ARCHITECTURE

2.1 The 6-Layer Architecture Stack

LAYER 6 — TRUST & IDENTITY
KYC Engine AML Engine Sanctions Screening Trust Score Verified Badge System Fraud Detection
LAYER 5 — PRESENTATION
Web App (Next.js 14) Mobile iOS (React Native) Mobile Android (React Native) Seller Dashboard Admin Portal Developer Docs Bank Portal
LAYER 4 — API GATEWAY
REST APIs GraphQL WebSocket Real-time Webhooks Rate Limiting API Keys OpenAPI 3.0 Docs
LAYER 3 — CORE ENGINES (12 Business Engines)
Marketplace Trade Finance Logistics Customs Compliance Payment Insurance AI/ML Escrow Notification Reporting
LAYER 2 — DATA LAYER
Firestore (real-time NoSQL) PostgreSQL (ACID transactions) Redis (cache) BigQuery (analytics) Elasticsearch (search) Firebase Storage
LAYER 1 — INFRASTRUCTURE
Google Cloud Platform Firebase Cloud Run Kubernetes (GKE) Cloud CDN Pub/Sub Cloud Armor (DDoS) Multi-region: US, EU, APAC, ME, Africa

2.2 Technology Stack

Layer	Technology	Purpose
Frontend Web	Next.js 14, TypeScript, Tailwind CSS, shadcn/ui	SSR web app, SEO-optimized, production-grade
Mobile (iOS+Android)	React Native + Expo SDK 51	Single codebase for both app stores
State Management	Zustand + TanStack Query v5	Global + server state, optimistic updates
Marketplace Search	Elasticsearch / Typesense	50M+ product search, <50ms response
Real-time	Firebase Firestore + Socket.io WebSocket	Live deal rooms, tracking, price feeds
Backend APIs	Node.js 20, Firebase Cloud Functions v2	Serverless, auto-scaling REST endpoints
Primary DB	Firestore (mock Phase 1 → production)	Real-time, globally distributed NoSQL
Transactional DB	PostgreSQL 16 (Cloud SQL)	ACID compliance for payments, contracts, escrow
Cache Layer	Redis (Upstash serverless)	FX rates (30s TTL), sessions, hot marketplace data
Analytics Warehouse	Google BigQuery	Trade reports, credit scoring, market intelligence
AI / ML Platform	Vertex AI + Gemini + custom TensorFlow models	KYC, risk scoring, price intelligence, matching
Payments	Custom engine + Stripe Connect + SWIFT API	Multi-currency, 150+ currencies, escrow
Monitoring	Google Cloud Monitoring + Sentry + Datadog	99.99% uptime SLA, error tracking, APM

2.3 The 22 Firestore Collections (Multi-Tenant from Day 1)

Every collection carries a `tenant_id` field. Multi-tenancy enforced at database level. This decision cannot be retrofitted later — it must be implemented before writing any module code.

Collection	Key Fields	Purpose
<code>/sellers</code>	<code>sellerId</code> , <code>company</code> , <code>country</code> , <code>products[]</code> , <code>trustScore</code> , <code>kycStatus</code> , <code>certifications[]</code> , <code>tier</code>	All verified sellers — the supply side of the marketplace
<code>/buyers</code>	<code>buyerId</code> , <code>company</code> , <code>country</code> , <code>industry</code> , <code>creditScore</code> , <code>kycStatus</code> , <code>purchaseHistory[]</code> , <code>creditLimit</code>	All verified buyers — the demand side
<code>/listings</code>	<code>listingId</code> , <code>sellerId</code> , <code>title</code> , <code>hsCode</code> , <code>category</code> , <code>price</code> , <code>currency</code> , <code>moq</code> , <code>leadTime</code> , <code>countries[]</code>	All marketplace product listings
<code>/rfqs</code>	<code>rfqId</code> , <code>buyerId</code> , <code>listingIds[]</code> , <code>quantity</code> , <code>targetPrice</code> , <code>incoterm</code> , <code>deadline</code> , <code>status</code> , <code>quotes[]</code>	Buyer RFQ requests sent to sellers
<code>/quotes</code>	<code>quoteId</code> , <code>rfqId</code> , <code>sellerId</code> , <code>price</code> , <code>currency</code> , <code>validity</code> , <code>terms</code> , <code>status</code>	Seller quote responses to RFQs
<code>/deal_rooms</code>	<code>dealId</code> , <code>buyerId</code> , <code>sellerId</code> , <code>messages[]</code> , <code>documents[]</code> , <code>status</code> , <code>agreedTerms{}</code> , <code>timeline</code>	Private per-trade negotiation spaces
<code>/trades</code>	<code>tradeId</code> , <code>dealId</code> , <code>value</code> , <code>currency</code> , <code>incoterm</code> , <code>status</code> , <code>milestones[]</code> , <code>disputeId</code>	All active and historical trades
<code>/contracts</code>	<code>contractId</code> , <code>tradeId</code> , <code>terms{}</code> , <code>signatures[]</code> , <code>smartContractHash</code> , <code>status</code> , <code>jurisdiction</code>	Smart contracts and legal agreements
<code>/escrow</code>	<code>escrowId</code> , <code>tradeId</code> , <code>buyerId</code> , <code>sellerId</code> , <code>amount</code> , <code>currency</code> , <code>status</code> , <code>releaseConditions[]</code>	Per-trade escrow accounts
<code>/payments</code>	<code>paymentId</code> , <code>tradeId</code> , <code>type</code> , <code>amount</code> , <code>currency</code> , <code>rails</code> , <code>status</code> , <code>settlementDate</code> , <code>fees{}</code>	All payment and settlement records
<code>/shipments</code>	<code>shipmentId</code> , <code>tradeId</code> , <code>carrier</code> , <code>mode</code> , <code>origin</code> , <code>destination</code> , <code>milestones[]</code> , <code>eta</code> , <code>blNumber</code>	Freight and logistics tracking
<code>/insurance</code>	<code>insuranceId</code> , <code>shipmentId</code> , <code>type</code> , <code>coverage</code> , <code>premium</code> , <code>provider</code> , <code>claimStatus</code> , <code>documents[]</code>	Cargo and trade insurance policies
<code>/customs</code>	<code>customsId</code> , <code>tradeId</code> , <code>hsCode</code> , <code>originCountry</code> , <code>destCountry</code> , <code>duties{}</code> , <code>filingStatus</code>	Customs declarations and duty records
<code>/kyc_cases</code>	<code>kycId</code> , <code>entityId</code> , <code>entityType</code> , <code>documents[]</code> , <code>amlFlags[]</code> , <code>sanctionMatches[]</code> , <code>riskScore</code>	KYC/AML compliance records
<code>/fx_rates</code>	<code>pair</code> , <code>bid</code> , <code>ask</code> , <code>mid</code> , <code>updatedAt</code> , <code>provider</code> , <code>volatility24h</code>	Live FX rates — 150+ currency pairs (30s cache)
<code>/hs_codes</code>	<code>hsCode</code> , <code>description</code> , <code>tariffRates{}</code> , <code>restrictions{}</code> , <code>ftaEligibility{}</code> , <code>lastUpdated</code>	Global HS code and tariff database
<code>/alerts</code>	<code>alertId</code> , <code>type</code> , <code>severity</code> , <code>entityId</code> , <code>message</code> , <code>module</code> , <code>isRead</code> , <code>createdAt</code>	Platform-wide alert system
<code>/users</code>	<code>userId</code> , <code>role</code> , <code>orgId</code> , <code>permissions[]</code> , <code>region</code> , <code>lastLogin</code> , <code>mfaEnabled</code>	User accounts and roles

/audit_logs	logId, userId, action, resource, before{}, after{}, ip, timestamp	Immutable complete audit trail
/disputes	disputeId, tradeId, raisedBy, reason, evidence[], status, arbitrator, resolution	Dispute and crisis management
/credit_scores	entityId, score, band, factors{}, tradingHistory{}, paymentHistory{}, lastUpdated	AI-generated credit scoring per entity
/market_data	commodity, price, currency, exchange, change24h, volume, updatedAt	Live commodity and market prices

SECTION 3: ALL 14 MODULES — STEP-BY-STEP WORKFLOW

Every module below describes exactly what it does, who uses it, the complete step-by-step workflow, every screen, and every API endpoint. This is the definitive build specification.

MODULE 1: Baalvion Marketplace (/marketplace) | P0 Critical | Phase 1

The front door. Every trade begins here. 50M+ listings. 4 industry verticals. 198 countries. Buyers find suppliers. Sellers reach global buyers. AI matches them. Nobody needs to go anywhere else to find a trading partner.

Who Uses It

- Buyers: search listings, compare prices, send RFQs, order samples
- Sellers: create listings, respond to RFQs, manage storefront
- Admins: moderate listings, manage categories, feature sellers

Step-by-Step Buyer Journey

1. Land on Marketplace homepage — see live platform stats, 4 industry vertical cards, featured listings, commodity price ticker
2. Search: keyword OR HS code OR plain language ('steel tubes for medical') — AI suggests correct HS code instantly
3. Filter results: MOQ, price, lead time, country of origin, certifications, seller tier (Basic/Verified/Premium/Platinum)
4. View listing detail: image gallery, pricing tiers by quantity, incoterms accepted, payment terms, certifications, factory photos, seller trust score breakdown (0-1000)
5. Check price intelligence panel: 'Market average for this commodity is \$X — this seller is Y% below/above market'
6. Take action: (a) Send RFQ to this seller, (b) Bulk RFQ to up to 20 sellers simultaneously, (c) Order 1-5 sample units, (d) Add to procurement shortlist, (e) Open Deal Room directly
7. Seller receives RFQ notification → opens inbox → views buyer requirements and verified badge → submits quote with AI-suggested pricing
8. Buyer views all quotes in comparison matrix → AI highlights best quote → awards to selected seller → Deal Room auto-created

Key Screens (10 total)

- /marketplace — Homepage with live stats, verticals, commodity ticker
- /marketplace/[category] — Infinite scroll listings grid with all filter options
- /marketplace/listing/[id] — Full listing detail, pricing tiers, price intelligence
- /marketplace/search — Advanced search, faceted filters, saved search alerts
- /marketplace/seller/[id] — Public seller storefront, all listings, reviews, trade stats
- /buyer/rfq — Sent RFQs, quote comparison matrix, award flow
- /seller/rfq — RFQ inbox, AI-priced quote builder
- /seller/listings/new — 5-step listing creation wizard
- /marketplace/tenders — Tender and reverse auction board
- /marketplace/prices — Commodity price intelligence hub

Key API Endpoints

- GET /api/listings — filter, sort, paginate (cursor-based, 50M+ scale)
- POST /api/rfq, GET /api/rfq/:id/quotes, POST /api/rfq/:id/award
- GET /api/search/listings — Typesense-powered with facets
- GET /api/ai/matches — AI buyer-seller recommendations

- GET /api/ai/price-intelligence — market price benchmarks
- POST /api/samples — sample order creation and tracking

MODULE 2: Deal Room & Smart Contracts (/deals) | P0 Critical | Phase 1

Replaces WhatsApp, email, DocuSign, and physical paper contracts. Every negotiation, every document, every signature, every escrow deposit — all inside one private, secure Deal Room per trade.

Step-by-Step Workflow

- Deal Room created automatically when buyer awards RFQ — both parties notified immediately
- Split-pane interface: real-time negotiation thread (left), deal summary panel (right) — WebSocket powered
- Counter-offer engine: propose price → counter → accept/reject — full version history with timestamps
- Term sheet co-editor: both parties simultaneously edit Incoterms 2020, payment terms, delivery schedule, quality specs
- Document vault: upload commercial invoice, packing list, Certificate of Origin, test reports, inspection reports
- Optional inspection booking: SGS, Bureau Veritas, or Intertek booked from inside Deal Room — report delivered to vault
- Smart contract auto-generated from agreed terms — ICC UCP 600 compliant, Incoterms 2020, jurisdiction clause, ICC/LCIA arbitration
- E-signature: both parties sign inside platform (finger draw on mobile / typed name) — certificate of signing issued immediately
- Escrow activation: buyer deposits trade value — seller sees 'Funds Confirmed' — production begins
- Milestone tracker: Contract Signed → Escrow Funded → Production → Quality Check → Shipped → Customs → Delivered → Payment Released
- Each milestone requires evidence upload before escrow releases — carrier API confirms delivery for final release

Key Screens

- /deals — Deal pipeline Kanban by stage
- /deals/[id] — Live deal room with negotiation thread and term sheet
- /deals/[id]/contract — Smart contract preview and e-signature
- /deals/[id]/documents — Document vault with version control
- /deals/[id]/escrow — Escrow deposit, milestone tracking, partial releases
- /deals/[id]/timeline — Visual milestone tracker with evidence slots

MODULE 3: Dashboard & Command Center (/dashboard) | P0 Critical | Phase 1

Nine completely different dashboards — one per user type. Every role sees only the data relevant to their function. The world map shows live trade flow arcs. KPIs update in real-time via WebSocket.

Portal	Key Dashboard Elements
Buyer	Active deals pipeline (Kanban), spend analytics by category/country/month, pending RFQs, top suppliers, FX watchlist, world map of shipments, procurement approval workflow
Seller	GMV tracker (daily/weekly/monthly/YTD), RFQ inbox widget, listing performance funnel, trust score breakdown, payout history and upcoming settlements, early payment CTA

Bank Officer	L/C issuance queue, credit utilisation per client, correspondent bank network, regulatory report generator, revenue earned per trade financed
Compliance Admin	KYC case queue heatmap, AML alert feed, sanctions hit rate, country risk matrix, audit trail explorer, pending escalations
Super Admin	Real-time platform KPIs (GMV/hr, users online), all services health grid (green/amber/red), global revenue by module, user growth, kill switch access
Government/Sovereign	Trade flow by commodity and corridor for their country, national export statistics, regulatory oversight tools, crisis intervention controls
Agent	Seller portfolio GMV table, commission tracker (0.5-1% of first-year GMV), leaderboard ranking, training completion, new seller onboarding tool
Developer	API usage by endpoint, P99 latency charts, error rate, quota burn rate, webhook delivery success, sandbox environment status
White-Label Admin	Tenant GMV, user count, module adoption, revenue sharing owed, brand configuration panel, tenant settings

MODULE 4: Buyer & Seller Onboarding (/onboard) | P0 Critical | Phase 1

Every participant is verified before they can trade. No anonymous buyers. No unverified sellers. KYC is not a formality — it is the foundation of the trust that enables billion-dollar trades between strangers.

Buyer Onboarding — 5 Steps

20. Account creation: email, password, country, business type
21. Company details: legal name, registration number, address, website, employee count, annual purchase volume
22. Document upload: company registration, director ID, bank statement, VAT/GST registration
23. AI KYC processing: document authenticity check, OFAC/UN/EU/UK/AU sanctions screening, PEP check, adverse media scan — under 2 hours
24. KYC approved: Verified Buyer badge, credit score (0-1000), platform tour, first 10 RFQs free

Seller Onboarding — 6 Steps

25. Account + company details (same as buyer plus product categories and annual export volume)
26. Document upload (same as buyer plus product certifications, factory audit report, export license)
27. Certification verification: AI reads certificate number, validates against issuing body, checks expiry
28. AI KYC + factory photo validation: detects stock images vs real factory photos — flags if stock image detected
29. Credit score and seller tier assignment: Basic (grey) → Verified (blue) → Premium (gold) → Platinum (platinum)
30. Seller badge ceremony — platform tour — listing creation wizard triggered immediately

MODULE 5: Compliance Hub (/compliance) | P0 Critical | Phase 1

Every entity is screened continuously — not once. Every trade triggers fresh sanctions screening. Every completed or disputed trade recalculates the trust score. The Compliance Hub is the platform's immune system.

Continuous Compliance Workflow

31. New entity submits documents → AI reads and validates: document type, expiry dates, identity fields extracted
32. Biometric verification: selfie matched against ID photo, liveness check (blink detection on mobile)

33. Sanctions screening: name, aliases, and associated companies checked against OFAC, UN, EU, UK, AU lists
34. AML risk assessment: business type + country + trade volumes vs FATF country risk grades
35. PEP screening: all directors and beneficial owners checked against global PEP database
36. Adverse media: news database scan for fraud, money laundering, corruption mentions in any language
37. Trust score calculated: 0-1000 composite across 6 factors — KYC tier, trade history, dispute rate, response rate, certifications, payment behaviour
38. Decision: Approved (badge issued) / Pending (info requested) / Rejected (with reason) / Escalated (senior officer)
39. Ongoing: every trade re-screens sanctions. Trust score recalculated after every trade event.

Anti-Fraud Systems

- Listing fraud detection: AI scans new listings for copied images, fake specs, price manipulation
- Identity spoofing: video selfie at onboarding, device fingerprinting on every login
- Shipping fraud: escrow only released after carrier API confirms delivery — not seller declaration
- Collusion detection: ML flags buyer-seller pairs transacting in unusual patterns
- Blacklist database: permanently banned entities — no re-registration, shared platform-wide

MODULE 6: Finance Module (/finance) | P0 Critical | Phase 2

Trade finance is the largest unmet need in global trade. Small exporters in India or Nigeria cannot access Letters of Credit from their local bank. Baalvion brings the full suite inside the platform — same day, for any verified trader.

Letters of Credit (L/C) — Workflow

40. Buyer applies for L/C in platform — specifies value, seller, shipment deadline, required documents
41. Baalvion routes to partnered bank (HSBC, Citi, Emirates NBD) — digital issuance same day vs 5-10 days paper
42. Seller sees L/C confirmed in Deal Room — begins production
43. Seller ships and uploads documents (Invoice, B/L, Packing List, CoO, Inspection Report)
44. Bank reviews documents — approves — payment released to seller

Other Finance Products

- Bank Guarantees: Performance Bond, Advance Payment Guarantee, Bid Bond
- Invoice Finance: upload approved invoice → funded up to 90% within 24 hours
- Trade BNPL: verified buyers pay in 30/60/90 days — seller paid immediately — Baalvion carries risk
- FX Engine: 150+ pairs, 30s live rates, rate lock (15 mins), forward contracts, multi-currency wallet
- Escrow Engine: per-trade escrow with milestone-based release — float interest is a revenue stream

MODULES 7 & 8: Customs & Logistics (/customs, /logistics) | P1 High | Phase 2

Customs Workflow

45. Describe product in plain language → AI HS code classifier returns top 3 suggestions with confidence score
46. Tariff calculator: HS code + origin + destination → full landed cost (import duty, VAT, excise, anti-dumping, clearance fee)
47. Export control check: dual-use goods flagged, prohibited country enforcement, export license requirement

48. Customs declaration auto-filled from trade data → country-specific form generated (US CBP, EU SAD, India Shipping Bill, UAE, China GACC) → submitted via API
49. Certificate of Origin auto-generated, e-stamped, submitted to chamber of commerce
50. Free Trade Zone routing: Jebel Ali, Singapore FTZ, Shanghai FTZ duty-free re-export auto-suggested

Logistics Workflow

51. Freight booking: enter origin, destination, cargo details → rate comparison across Maersk, MSC, CMA CGM, DHL, 50+ carriers
52. Select mode: sea/air/road/rail with transit time, price, and CO2 footprint per option
53. Book in one click — booking confirmation and reference returned
54. Digital Bill of Lading: created in platform, title transferred electronically, surrendered digitally
55. Live tracking: shipment on world map, vessel position updated in real-time, milestone timeline
56. Exception management: port congestion, weather delay, carrier issue — ETA impact + escalation flow
57. Carbon footprint calculated per shipment — offset purchase option — ESG report integration

MODULE 9: Global Payment Rails (/payments) | P0 Critical | Phase 2

Baalvion owns the money layer. Escrow holds funds. Settlement releases them only when milestones are confirmed. 9 payment rails. 150+ currencies. 24-hour settlement. This is the trust infrastructure.

Payment Rails Supported

- International: SWIFT wire transfer, SEPA (Europe), ACH (USA)
- India: UPI, NEFT, RTGS (via Authorized Dealer bank)
- Brazil: Pix instant payment
- Singapore: PayNow
- Africa: M-Pesa, MTN Mobile Money (via Flutterwave/Paystack)
- Latin America: PSE (Colombia), SPEI (Mexico), MercadoPago
- China: UnionPay, cross-border RMB (via Ant Group API)

Settlement Workflow

58. Buyer deposits trade value into Baalvion escrow
59. Escrow confirmed — seller begins production
60. Platform earns float interest on escrow balances — revenue stream
61. Each milestone met → partial or full escrow release
62. Carrier API confirms delivery → final release
63. Seller receives funds in chosen currency within 24 hours
64. Nightly reconciliation engine: all transactions auto-reconciled against Firestore records

MODULE 10: Insurance (/insurance) | P1 High | Phase 2

Insurance Products

- Cargo Insurance: instant quote (cargo type, value, route, Incoterms) → compare Lloyd's of London and Allianz → purchase in minutes
- Trade Credit Insurance: covers buyer default risk — underwriter sets coverage limit based on buyer credit score
- Parametric Insurance: auto-pays out when ship delayed >14 days — no claims process, no evidence required — triggered by carrier API data

Claims Workflow

65. File claim from trade page — upload photo/video evidence
66. Parametric claims auto-settled when trigger confirmed by carrier data
67. Non-parametric: claim submitted to underwriter via API — decision within policy SLA
68. Settlement paid into Baalvion wallet — available for next trade

MODULE 11: AI Intelligence Hub (/analytics) | P1 High | Phase 3

After 2 years of real trades, Baalvion holds the world's most valuable global trade dataset. The AI Hub turns that data into unfair advantage for every user on the platform.

- Buyer-Seller Matching: AI match score 0-100, 5-factor breakdown, 'Why this match?' explanation
- Price Intelligence: 90-day commodity benchmarks from real transactions — 'Is this price fair?' tool
- Baalvion Trade Score (BTS): 0-1000 creditworthiness score — like FICO but for global trade — factors: payment history, trade volume, dispute rate, verification tier, tenure, counterparty quality
- Supplier Risk Early Warning: detect seller approaching default — 30-day early warning signal
- Demand Forecasting: tell sellers what buyers will want in the next 90 days
- Natural Language Trade Assistant: 'Find 3 steel suppliers in Turkey, 500MT HRC, ship by March' → instant results with match scores
- Baalvion Trade Intelligence API (BTI): sell anonymised trade flow data to banks and hedge funds at \$50K-\$500K/year

MODULE 12: Admin Control Center (/admin) | P0 Critical | Phase 1

Super Admin Controls

- Kill switch: per-country suspension, per-module disable, per-user emergency ban — approval workflow before execution
- User management: search all users, view KYC, change role, freeze account, view device sessions
- Platform fee management: set transaction fees per module, per tier, per tenant
- Content moderation: flagged listing queue, bulk approve/reject/remove

Compliance Admin Controls

- KYC case queue with bulk review tools
- AML alert feed with real-time incoming cases
- Sanctions screening live results
- Audit log explorer: every action searchable, filterable, exportable

Government Sovereign Controls

- National trade flow statistics by commodity and corridor
- Regulatory oversight tools — view trades involving their country's entities
- Crisis intervention: request trade freeze, account investigation

MODULE 13: Crisis & Dispute Center (/crisis) | P1 High | Phase 3

3-Tier Dispute Resolution

Tier	Timeframe	How It Works	Outcome
------	-----------	--------------	---------

Tier 1 — AI	0 – 48 hours	AI reviews shipping proof, delivery confirmation, photo evidence. Auto-decision for clear-cut cases.	Auto-refund or auto-release based on evidence weight
Tier 2 — Human Mediator	48hr – 14 days	Baalvion trained mediator reviews all evidence and makes binding platform decision.	Winning party compensated. Loser receives trust score strike.
Tier 3 — ICC Arbitration	14 – 60 days	International Chamber of Commerce arbitration for disputes above \$100,000.	Legally binding international arbitration award

- Seller has 72-hour mandatory response deadline — failure = automatic ruling for buyer
- Strike system: 3 strikes → 30-day suspension. 5 strikes → permanent ban + industry blacklist

MODULE 14: Reports & Developer Platform (/reports, /developer) | P2 Medium | Phase 4

Reports

- Report builder: drag-drop dimension and metric picker, date range filters, live preview before generation
- Pre-built library: Trade Volume by Corridor, Compliance Hit Rate, GMV by Industry, Agent Performance
- BTI premium reports: anonymised market data, price benchmarking — Enterprise tier
- ESG reports: Scope 3 supply chain emissions, SDG alignment, conflict minerals (OECD 3TG)
- Export: PDF, CSV, Excel — scheduled delivery by email

Developer Platform

- Interactive API docs: Swagger/OpenAPI 3.0, live try-it-now for all 60+ endpoints
- Sandbox environment: realistic test data, mock KYC, mock payment flows
- Webhook management: register endpoints, select from 40+ event types, test delivery, event log
- SDK downloads: JavaScript, Python, Go with quickstart guides
- Developer analytics: API usage, P99 latency, error rate, quota burn rate

SECTION 4: THE 9 USER PORTALS

Portal	Primary Need	Their Baalvion Experience
Buyers	Find reliable global suppliers and receive goods as ordered	Marketplace browsing, bulk RFQ, deal room, escrow, logistics tracking, payment history, dispute filing, spend analytics, procurement approval workflow
Sellers	Reach global buyers and get paid reliably and fast	Listing creation, RFQ inbox, quote builder, deal room, document vault, logistics booking, invoice finance, payout management, trust score dashboard
Banks	Finance trades and manage credit risk	L/C issuance queue, credit line management, correspondent bank network, revenue dashboard, regulatory reporting
Compliance Teams	Monitor AML, sanctions, and fraud	KYC case queue, AML alert feed, sanctions screening dashboard, trust score management, fraud detection, blacklist management, audit trail
Super Admins	Operate and protect the platform	Platform KPIs, kill switch, user management, fee management, content moderation, platform constitution, announcement system
Governments	Trade statistics and regulatory oversight	National trade flow dashboard, export statistics, regulatory tools, crisis management, Baalvion transparency reports
Agents	Onboard sellers and earn commission	Seller portfolio, commission tracker (0.5-1% GMV), leaderboard, training modules, seller onboarding tool
Developers	Build integrations on top of Baalvion	API docs, sandbox, API keys, webhooks, SDK, usage analytics, webhook event log
White-Label Admins	Operate a branded Baalvion tenant	Brand config, module toggles, tenant GMV, user management, fee overrides, revenue sharing dashboard

SECTION 5: BUSINESS MODEL — 11 REVENUE STREAMS

Revenue Stream	Rate	Phase Live	Annual Potential at Scale
Marketplace Transaction Fees	0.05–0.15% of deal value	Phase 2	\$5–15 Billion/year at \$10T volume
Trade Finance Origination	0.5–2.0% of financed amount	Phase 2	\$3–8 Billion/year
FX Spread Revenue	0.1–0.3% of conversions	Phase 2	\$2–6 Billion/year
Subscription Plans (SaaS)	\$99 / \$499 / \$4,999/month	Phase 1	\$500M–2 Billion/year
Logistics Booking Commission	2–5% of freight value	Phase 3	\$1–3 Billion/year
Insurance Commission	8–12% of premium	Phase 3	\$500M–1 Billion/year
Customs Filing Fees	\$25–200 per declaration	Phase 3	\$200M–500M/year
Compliance-as-a-Service API	\$0.50–5.00 per KYC check	Phase 4	\$300M–800M/year
Trade Intelligence Data API (BTI)	\$50K–\$500K/year per subscription	Phase 4	\$500M–2 Billion/year
Developer API Platform	Usage-based billing	Phase 5	\$200M–500M/year
White-Label Platform License	\$100K–2M/year enterprise	Phase 5	\$500M–2 Billion/year

Total Addressable Revenue at Scale: \$12 – \$38 Billion Per Year | At 0.1% market penetration: \$1.2 – 3.8 Billion/year

SECTION 6: 6-PHASE BUILD ROADMAP

Phase	Timeline	What We Build	Key Milestone	MRR
Phase 1	Months 1–3	App shell, Marketplace, Dashboard, Trade Creation, KYC/Compliance, Admin Center, Mock APIs, Firestore seed (all 22 collections), design system, component library	50 verified sellers, 100 buyers, first live trade on platform	\$0
Phase 2	Months 4–6	Finance Module (L/C, BG, Invoice Finance, BNPL), Compliance Hub full, Deal Room complete, Customs Engine, Payment Rails, FX Engine, banking partner integrations	\$10M volume, 500 active users, first L/C issued on platform	\$50K
Phase 3	Months 7–9	Logistics Layer, Cargo Insurance, Customs auto-filing, React Native iOS + Android mobile app, 10 trade corridors active	Mobile app live, 10 corridors, \$100M volume processed	\$200K
Phase 4	Months 10–13	AI Hub, Credit Scoring, Price Intelligence, Fraud Detection, Natural Language Assistant, Demand Forecasting, 50-country expansion	AI matching live, \$1B annual volume milestone, BTI API launched	\$1M
Phase 5	Months 14–16	Developer Platform, Public API, 100 countries, government partnerships, white-label product, first sovereign integration	First government integration, 1,000 verified organisations	\$3M
Phase 6	Months 17–24	198-country full coverage, enterprise white-label at scale, IPO-ready metrics, Series B fundraise preparation	\$10B+ annualized volume, 10,000 organisations, Series B closed	\$10M+

7 Non-Negotiable Decisions — Before Writing Any Code

#	Decision	Answer — Non-Negotiable
D1	Multi-tenant vs Single-tenant	Multi-tenant from Day 1. tenant_id on every DB table. Cannot be retrofitted. Banks and govts require this.
D2	B2B only vs B2B + B2C	B2B ONLY for first 3 years. Consumer trade adds 10x compliance burden with 0.1x revenue per transaction.
D3	Open vs Curated marketplace	Curated: every seller manually approved for first 1,000 sellers. Opens to self-serve at 1,000+.
D4	Own the money vs Pass-through	Own the money. Escrow is the trust foundation. Get payment licenses early. Non-negotiable.
D5	Build vs Buy compliance tools	Buy first: ComplyAdvantage, Onfido. Don't build KYC from scratch. Integrate, then build proprietary layer on top.
D6	Monolith vs Microservices	Modular monolith Phase 1. Migrate to microservices module by module from Phase 3. Never big bang.
D7	Geographic launch sequence	One corridor only: India–UAE (\$80B annual trade). Dominate it. Then corridor 2. Never launch 10 simultaneously.

SECTION 7: THE 350-PROMPT PRODUCTION BUILD PLAN

350 production-grade prompts builds the complete Baalvion codebase. After all 350 prompts: the frontend is permanent — no frontend team ever touches it again. The backend team's only job is replacing mock API responses with real vendor integrations, one by one, without touching a single screen.

Phase / Module	What Gets Built	Prompts	Range
Phase 0	Firestore schema (22 collections), Auth (9 roles, JWT, route guards), Design system, 80+ UI components, 60+ API endpoint stubs, WebSocket rooms, Notification engine, Multi-tenancy layer, Redis cache, PDF templates, Typesense search setup	15 prompts	P-01 → P-15
Module 1: Marketplace	Homepage, listings grid (infinite scroll, all states), advanced search + facets, HS code AI lookup, AI matching panel, bulk RFQ engine, buyer RFQ management + quote matrix, seller RFQ inbox + AI quote builder, listing wizard (5 steps), listing management + analytics, seller storefront, sample order system, tender/auction, price intelligence hub, procurement hub + approval workflow, verified reviews, admin moderation, localisation (6 languages + RTL), SEO/SSR, complete API layer	35 prompts	P-16 → P-50
Module 2: Deal Room	Deal pipeline + list view, live negotiation thread (WebSocket), counter-offer engine, term sheet co-editor, in-deal messaging, document vault, inspection booking, smart contract generator, e-signature flow, legal clause overlay, escrow deposit/release, milestone tracker, multi-party rooms, deal cloning, deal analytics, video call integration, deal archive, all notifications, mobile variant, complete API layer	25 prompts	P-51 → P-75
Module 3: Dashboards	All 9 role-specific dashboards, world trade flow map, live commodity ticker, FX watchlist, alert centre, dashboard customisation, PDF export, mobile variant, BigQuery analytics API layer	20 prompts	P-76 → P-95
Module 4: Onboarding	Buyer KYC wizard, seller KYC wizard (6 steps), organisation management, bank onboarding, agent application, government registration, developer registration, interactive platform tour, subscription selection at onboarding, 14-email sequence builder, complete API layer	20 prompts	P-96 → P-115
Module 5: Compliance Hub	KYC case management (5 prompts), AML monitoring (4 prompts), sanctions screening (3 prompts), trust score system (3 prompts), fraud detection (3 prompts), blacklist + certification verification (3 prompts), export control (3 prompts), compliance reporting, complete API layer	25 prompts	P-116 → P-140
Module 6: Finance	Letters of Credit (5 prompts), Bank Guarantees (4 prompts), Invoice Finance (4 prompts), Trade BNPL (3 prompts), FX engine + rate lock (5 prompts), multi-currency wallet (4 prompts), bank officer portal (4 prompts), treasury display, complete finance API layer	30 prompts	P-141 → P-170
Module 7: Payments	SWIFT/SEPA/ACH flows, 6 local rail payment UIs (UPI, Pix, PayNow, M-Pesa, PSE, SPEI), payment status tracking, settlement engine UI, payout management, subscription billing, dispute payment hold, accounting export (QuickBooks/Xero/SAP/Tally), complete API layer	25 prompts	P-171 → P-195

Module 8: Customs	AI HS classifier (3 prompts), tariff calculator (3 prompts), declaration filing 5 country templates (3 prompts), Certificate of Origin (2 prompts), export control flags (3 prompts), country risk map (2 prompts), FTZ integration, customs document management, complete API layer	20 prompts	P-196 → P-215
Module 9: Logistics	Freight booking + carrier comparison (4 prompts), shipment creation (3 prompts), live tracking + world map (3 prompts), digital Bill of Lading (3 prompts), trade document generation (2 prompts), carbon footprint tracker (2 prompts), freight forwarder portal, exception management, complete API layer	20 prompts	P-216 → P-235
Module 10: Insurance	Cargo quote + purchase flow (3 prompts), trade credit insurance (2 prompts), parametric insurance (1 prompt), policy management (2 prompts), claims filing + tracker (2 prompts), insurance admin view, complete API layer	13 prompts	P-236 → P-248
Module 11: AI Hub	Buyer-seller matching (3 prompts), price intelligence hub (3 prompts), BTS credit scoring (3 prompts), supplier risk warning (3 prompts), demand forecasting (3 prompts), NL trade assistant (2 prompts), fraud pattern visualisation, AI model dashboard, complete API layer	20 prompts	P-249 → P-268
Module 12: Reports	Report builder (3 prompts), pre-built library (3 prompts), BTI premium reports (2 prompts), ESG reporting (2 prompts), export engine PDF/CSV/Excel (2 prompts), report viewer, custom KPI builder, complete API layer	15 prompts	P-269 → P-283
Module 13: Admin	Super admin command center, kill switch panel, user management, audit log explorer, fee management, content moderation, announcement system, sovereign portal, agent management, white-label tenant management, complete API layer	15 prompts	P-284 → P-298
Module 14: Crisis + Disputes	Dispute filing flow, Tier 1 AI resolution UI, Tier 2 human mediator, Tier 3 ICC arbitration, SLA countdown timers, strike system, dispute history, crisis dashboard, platform incident management, complete API layer	15 prompts	P-299 → P-313
Module 15: Developer Platform	Dev portal hub, interactive API docs (all 60+ endpoints), sandbox environment, webhook management, SDK download page, developer analytics, complete API layer	10 prompts	P-314 → P-323
Mobile: iOS + Android	Auth + biometric (2 prompts), onboarding splash, React Navigation setup, marketplace all screens (3 prompts), deal room mobile (3 prompts), e-signature + escrow mobile (2 prompts), all 9 dashboards mobile (3 prompts), payments + FX (2 prompts), KYC camera + liveness (2 prompts), logistics map (2 prompts), AI assistant, notifications, profile, settings, offline mode, RTL Arabic/Urdu, App Store assets, Play Store assets, EAS Build config, mobile API layer, E2E test (Detox)	50 prompts	P-324 → P-373
Security + Compliance	CSP headers, CORS lockdown, API rate limiting (Redis), input sanitisation, GDPR cookie consent, GDPR data export, GDPR data deletion, PDPA compliance, PIPL compliance, privacy policy version management, 2FA enforcement, session management, pen test checklist UI, DDoS protection (Cloud Armor), data encryption display, vulnerability disclosure page, security audit log	17 prompts	P-374 → P-390
Testing + CI/CD + Launch	Full trade journey E2E integration test, unit test suite (80%+ coverage), component tests (React Testing Library), Playwright E2E (10 critical journeys), k6 load test (1000 concurrent), GitHub Actions CI/CD pipeline, GCP multi-region config (US/EU/APAC/ME), Sentry +	10 prompts	P-391 → P-400

	Cloud Monitoring setup, staging vs production split, launch readiness checklist		
--	---	--	--

TOTAL: 350 production prompts. 200+ screens. 60+ API endpoints. 9 user portals. iOS + Android. After P-400: frontend team needed = ZERO.

SECTION 8: GO-TO-MARKET STRATEGY

8.1 Cold Start Strategy — Supply-First Launch

The hardest problem in marketplaces: who comes first, the buyer or the seller? Strategy: recruit 500 verified sellers BEFORE opening to buyers. Day 1 buyers see a full marketplace — not an empty one.

Strategy	How It Works	Target Outcome
Supply-First Launch	Recruit 500 verified sellers before opening to buyers. White-glove onboarding. Free subscription for first 6 months.	Day 1 buyers see a full marketplace — not an empty one
One Corridor First	Launch ONLY India-UAE corridor (\$80B annual trade). Dominate it. Then expand.	\$10M volume in first corridor before adding corridor 2
Trade Association Deals	Partner with FIEO (India), CCIA (China), CBI (Netherlands) — get member lists as founding sellers	1,000 verified sellers from one partnership deal
Anchor Buyer Strategy	Sign 5 Fortune 500 procurement teams as launch partners with committed purchase volume	500 SME sellers rush to list to access those anchor buyers
Government Endorsement	Approach DGFT India, UK DIT — become official digital trade platform for national exporters	Government endorsement = instant national credibility
Invite-Only Phase 1	Waitlist — 'apply for early access' — scarcity creates demand and quality filtering	Buzz + press coverage before launch. Quality users only.
10,000 Agent Network	Local trade agents in 198 countries onboard sellers, earn 0.5-1% of first-year GMV	Human-powered acquisition where internet-first won't reach

8.2 Funding Roadmap

Round	Amount	Timing	Use of Funds
Pre-Seed	\$500K – \$1M	Month 0 — Now	350-prompt build, legal entity, first license applications, founding team (CTO, Legal Counsel, Compliance Expert)
Seed	\$3M – \$8M	Month 6	Banking partnerships, compliance team, Onfido/ComplyAdvantage contracts, 10-country launch, marketing
Series A	\$20M – \$50M	Month 12	Mobile app, logistics APIs, AI team, 50-country expansion, payment licenses across key markets
Series B	\$100M – \$200M	Month 24	198-country, white-label product, Trade Intelligence API, physical inspection offices
Series C+	\$500M+	Month 36	Acquisitions, IPO preparation, sovereign integrations, global HQ

SECTION 9: TRUST, SAFETY & LEGAL REQUIREMENTS

9.1 Licenses Required — Apply on Day 1 (12-24 Month Queue)

A single regulatory violation can shut Baalvion down in an entire country. Every license below must be applied for immediately. The licensing queue — not the technology — is the constraint on launch timeline.

License	Required In	Regulator	Risk If Missing
Payment Institution License	EU, UK, Singapore, UAE	FCA, MAS, CBUAE	Cannot process payments — platform dead
Money Transmitter License (MTL)	USA — all 50 states	FinCEN + state regulators	Illegal to operate — criminal liability
Banking Aggregator License	India	Reserve Bank of India	Cannot offer trade finance or escrow
AML/CFT Registration	All active markets	National FIUs	Fines, suspension, criminal charges
E-Commerce Marketplace License	China, Indonesia, UAE	MOFCOM, MCI, DED	Cannot list products or take commission
Insurance Intermediary License	EU, UK, US, Australia	Local insurance regulators	Cannot sell insurance products
Customs Broker License	US, EU, India, Australia	CBP, HMRC, CBIC	Cannot file customs declarations
Financial Services License (FSL)	Singapore, HK, Australia	MAS, SFC, ASIC	Cannot offer L/C or trade finance
GDPR Data Processing Agreement	EU — mandatory	National Data Protection Authorities	EUR 20M or 4% global revenue fine

SECTION 10: KPIs & NORTH STAR METRICS

Metric	Phase 1 (M3)	Phase 3 (M9)	Phase 5 (M16)	Endgame
Verified Sellers	100	5,000	100,000	10M+
Verified Buyers	200	20,000	500,000	50M+
Monthly Trade Volume	\$10M	\$500M	\$10B	\$1T+
Countries Active	10	50	150	198
API Uptime SLA	99.5%	99.9%	99.99%	99.999%
KYC Approval Time	< 2 hours	< 30 minutes	< 5 minutes	< 1 min AI
L/C Issuance Time	Same day	< 4 hours	< 1 hour	< 10 minutes
Deal-to-Contract	< 3 days	< 24 hours	< 4 hours	< 1 hour
Monthly Revenue (MRR)	\$0	\$500K	\$5M	\$100M+
Net Promoter Score	Testing	> 45	> 60	> 70

BAALVION INDUSTRIES PRIVATE LIMITED

The world has never had a true Global Trade Operating System. Until Now.

198 countries. Every verified buyer. Every verified seller. Banking, logistics, customs, compliance, insurance — one platform. Zero reason to go anywhere else.

Document	Version	Classification
Baalvion Industries Pvt. Ltd. — Master Project Plan	v1.0 — 2025	CONFIDENTIAL — Internal Use Only
Consolidated from 5 source documents	Baalvion Industries Private Limited	All rights reserved